Can we better identify high-risk loans? Evidence from the German Credit Dataset

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This paper presents an empirical study of credit risk on the German Credit dataset using generalized linear models and resampling-based uncertainty assessment. The dataset contains 1,000 credit applicants with socio-economic and financial variables such as loan duration, credit amount, savings, employment duration, and demographic characteristics, together with a binary outcome indicating good or bad credit risk. Our objective is to build an interpretable model for predicting credit default and to compare several approaches to regularization and model validation in this setting. We first fit a full logistic regression model with one-hot encoded categorical predictors as a baseline. Based on p-values from the logistic regression summary, we then construct a reduced specification retaining only variables with the strongest statistical evidence, trading off parsimony against potential loss of predictive power. Next, we estimate an ℓ_1-regularized (Lasso) logistic regression on standardized predictors to perform automatic feature selection and control overfitting. For each model, we evaluate out-of-sample performance using a train–test split and metrics such as accuracy, recall, F1-score, and deviance, with particular emphasis on recall for the bad-credit class due to its higher decision cost. To quantify estimation uncertainty and stability, we apply bootstrap resampling to the logistic models and perform k-fold cross-validation to study bias–variance trade-offs in predictive performance. Finally, we analyze the empirical distributions of predicted risk scores for good and bad clients to assess separation and irreducible error. Our results show that a regularized Lasso logistic model achieves competitive overall accuracy while yielding a sparse, interpretable set of predictors and relatively stable performance across folds. However, the bad-class recall remains only moderate and exhibits non-negligible variability, indicating that even well-tuned linear models face intrinsic limitations in discriminating high-risk clients in this dataset.

I. Introduction

Credit scoring plays a central role in modern financial systems. Banks and lending institutions must decide whether to grant loans to applicants while managing the risk of default. Incorrect assessments have asymmetric costs: granting credit to a high-risk client may result in financial loss, whereas rejecting a reliable client may reduce revenue and harm customer relations. Because of this, credit risk modeling has become a classical application area for statistical learning methods.

The German Credit dataset provides a controlled environment for studying this problem. It contains information on 1,000 credit applicants together with a binary outcome indicating good or bad credit risk. The predictors include demographic characteristics, employment information, savings and checking account categories, loan characteristics, and other financial indicators. This structure makes the dataset suitable for evaluating methods that balance predictive performance with interpretability.

In this project, our goal is not only to predict credit default but also to understand how different statistical modeling choices affect performance, uncertainty, and interpretability. Rather than focusing on complex black-box models, we rely primarily on generalized linear models and resampling techniques that allow us to reason about coefficients, uncertainty, and model assumptions.

We begin with a full logistic regression model including all available predictors after appropriate preprocessing. We then construct a reduced specification by removing predictors with weak statistical evidence, aiming to obtain a more parsimonious model without large losses in predictive power. To address potential overfitting and automatic feature selection, we also estimate an ℓ_1-regularized (Lasso) logistic regression on standardized predictors. Throughout the analysis, we place particular emphasis on the bad-credit class, since misclassifying high-risk applicants is typically more costly than rejecting low-risk ones.

Model performance is evaluated with out-of-sample metrics such as accuracy, recall, and F1-score. To quantify uncertainty and assess stability, we use bootstrap resampling and k-fold cross-validation. Finally, we analyze the distributions of predicted

probabilities to investigate how well the models separate the two classes and to discuss the potential presence of irreducible error in this problem.

This leads to the central research question of this paper: *How well can interpretable logistic regression models predict credit risk on the German Credit dataset, and how do regularization and resampling techniques affect their performance, stability, and interpretability?*

Based on prior literature and intuition about credit scoring, we formulate the following hypotheses:

H1 (Model usefulness). *Logistic regression models achieve predictive performance clearly above a naive baseline (e.g., always predicting the majority class).*

H2 (Regularization benefit). *An ℓ_1-regularized (Lasso) logistic regression improves generalization performance compared to an unregularized model by reducing overfitting and removing weak predictors.*

H3 (Class imbalance effect). *Even when overall accuracy is high, recall for the bad-credit class remains noticeably lower, reflecting the inherent difficulty of detecting high-risk applicants.*

H4 (Stability). *Bootstrap and cross-validation reveal non-negligible variability in coefficients and performance metrics, indicating that conclusions based on a single train–test split may be misleading.*

H5 (Threshold trade-off). *Lowering the classification threshold increases recall for bad-credit applicants, but reduces overall accuracy by increasing the number of false positives.*

II. Data and Preprocessing

The analysis is based on the German Credit dataset, which contains information on 1,000 loan applicants together with a binary indicator of creditworthiness. The response variable classifies each applicant as either good credit risk or bad credit risk according to the bank's historical assessment. The predictors include both numerical and categorical variables describing financial behavior, credit history, and basic socio-demographic attributes. The dataset includes variables such as:

- Credit amount and loan duration
- Status of existing checking account
- Savings account category
- Employment duration
- Installment rate as a percentage of income
- Housing status and number of dependents
- Purpose of the loan (e.g., car, appliances)
- Existing credit at the bank
- Age in years

Most predictors are categorical with ordered or nominal levels. This structure makes the dataset suitable for interpretable statistical modeling, while still reflecting some of the complexity of real credit scoring systems.

The target variable is binary:

- 1 = good credit risk
- 0 = bad credit risk

Because good credit cases are more frequent, the dataset is moderately imbalanced. This motivates evaluating metrics beyond accuracy, with particular emphasis on recall for the bad-credit class.

To gain an initial understanding of the relationship between customer behavior and credit outcomes, we examined how the number of existing credits relates to default risk..

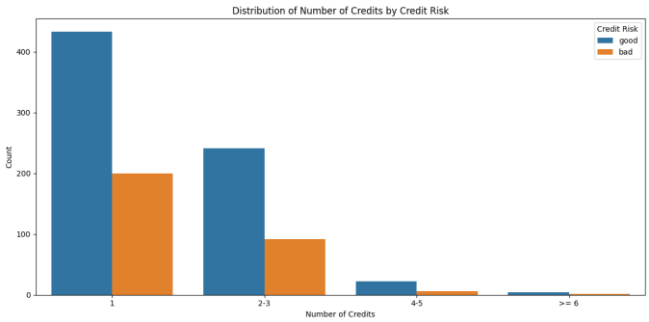

Fig. 1. Distribution of number of credits by credit risk.

A. Handling missing values

The dataset does not contain explicit missing values. However, several categorical variables include categories that implicitly encode missing or undefined situations (for example, "no checking account"). These categories were retained rather than imputed, since they may contain meaningful information about financial behavior.

B. Encoding of Predictors

Logistic regression requires numeric inputs. Therefore, all categorical predictors were transformed using one-hot encoding, creating indicator variables for each category while avoiding perfect multicollinearity through the standard reference-category approach. This allows the coefficients to be interpreted relative to a baseline level for each variable.

C. Scaling

For the unregularized logistic regression models, scaling is not strictly necessary because coefficients are estimated independently for each predictor. However, Lasso regularization applies a penalty based on coefficient magnitude, which is sensitive to the scale of variables.

Therefore: continuous predictors were standardized to zero mean and unit variance; and dummy variables were left as 0/1 indicators

This ensures that the regularization penalty treats predictors more comparably.

D. Train-Test Split

To evaluate out-of-sample performance, the dataset was randomly divided into: 75% training data for fitting the models and 25% test data for evaluation
The split was stratified with respect to the outcome variable to preserve the original class proportions in both sets.

III. METHODS

Our methodological strategy is designed to balance three goals: (i) predictive performance, (ii) interpretability, and (iii) assessment of statistical uncertainty. We therefore rely primarily on generalized linear models together with regularization and resampling techniques.

A. Logistic Regression

We first estimate a standard logistic regression model, where the probability of being a good credit risk is modeled as

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta},$$

with p_i denoting the probability of good credit for applicant i, and $\mathbf{x}_i$ representing the encoded predictor variables. Parameters are estimated via maximum likelihood.

This model provides: interpretable coefficients, statistical inference via standard errors and p-values, a baseline against which alternative specifications can be compared. We refer to this as the full model, as it initially includes all available predictors after preprocessing.

Fig. 2 highlights the predictors with the largest estimated effects in the full logistic model.

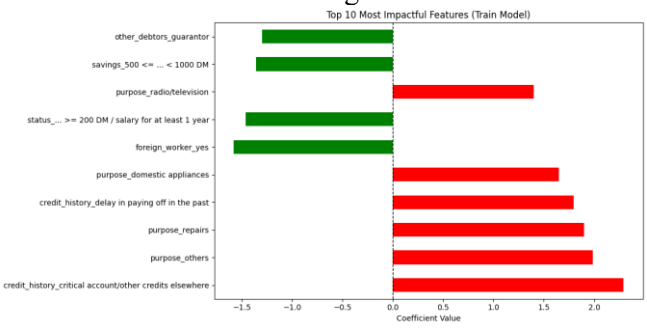

Fig. 2. Top ten logistic regression coefficients (absolute value).

B. Reduced Logistic Model (Model selection by p-values)

To examine whether a simpler model can achieve similar performance, we construct a reduced specification. Predictors whose coefficients display weak statistical evidence (high p-values in the logistic summary) are iteratively removed, while ensuring that potentially important variables are retained.

The motivation is twofold:

1. improve interpretability and parsimony
2. reduce variance and potential overfitting

The final reduced set is determined pragmatically, balancing statistical evidence and domain plausibility rather than relying on a purely automatic selection rule.

C. Lasso-Regularized Logistic Regression

While p-value–based selection relies on repeated hypothesis testing, Lasso regularization integrates variable selection directly into estimation. We therefore estimate an ℓ_1-penalized logistic model:

$$\text{logit}(p_i) = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \text{subject to } \lambda \sum_j |\beta_j|,$$

where λ controls the strength of the penalty. Larger values push more coefficients toward zero, potentially removing irrelevant predictors.

Because the penalty depends on coefficient scale, continuous predictors are standardized prior to fitting. The tuning parameter λ is selected using cross-validation on the training data.

This model allows us to evaluate H2, i.e., whether regularization improves generalization compared to the unregularized model.

D. Evaluation Metrics

Given the class imbalance and asymmetric decision costs, relying solely on accuracy can be misleading. We therefore report:

- Accuracy - overall fraction correctly classified
- Recall (Bad class) – proportion of bad credits correctly identified
- Precision and F1-score – harmonic balance between recall and precision
- Deviance / log-loss – probabilistic goodness-of-fit measure

Special emphasis is placed on recall for the bad-credit class, which relates to hypothesis H3.

E. Bootstrap Resampling

To assess uncertainty in estimates and predictions, we apply bootstrap resampling to the logistic models:

1. repeatedly sample (with replacement) from the training data
2. re-estimate the model each time
3. compute distributions of coefficients and performance metrics

This provides insight into: variability of estimated effects, stability of predicted probabilities, whether conclusions are robust to sampling fluctuations. Bootstrap results contribute directly to hypothesis H4 (stability).

Beyond point estimates, we are also interested in the sampling variability of our models. The bootstrap allows us to approximate how our results would change if we repeatedly observed new samples from the same

population. In our setting, each bootstrap replicate corresponds to re-running the entire modelling pipeline, thereby propagating uncertainty through feature selection and estimation.

F. Cross-Validation

To further evaluate generalization performance and reduce dependence on a single train–test split, we use k-fold cross-validation: the training set is divided into k folds, models are trained on $k-1$folds and validated on the remaining fold, the process repeats across folds and metrics are averaged. Cross-validation is also used to select the optimal Lasso penalty parameter.

Finally, we analyze the distribution of predicted probabilities for good and bad credit applicants, using density plots and histograms. This provides an intuitive view of **class separation** and highlights potential irreducible error when distributions overlap substantially.

IV. RESULTS

The dataset consists of 1,000 observations, of which roughly 70% correspond to good credit and 30% to bad credit. A naive majority-class classifier would therefore achieve approximately 70% accuracy, providing a useful baseline for comparison. Exploratory summaries indicate that clients classified as bad credit tend to have higher loan amounts and less favorable credit history categories, although there is substantial overlap across most predictors.

A. Full Logistic Regression

The full logistic regression model, including all 55 one-hot encoded predictors, achieved: Test accuracy: 0.72, Bad-class F1-score: 0.47 and Bad-class recall: 0.41
The classification report indicated noticeably lower recall for bad credit than good credit. Several credit-history and purpose categories showed strong associations with the outcome, but many coefficients exhibited wide confidence intervals, suggesting potential overfitting.

To understand classification behavior beyond aggregate metrics, we analyze the confusion matrix shown in Fig. 3.

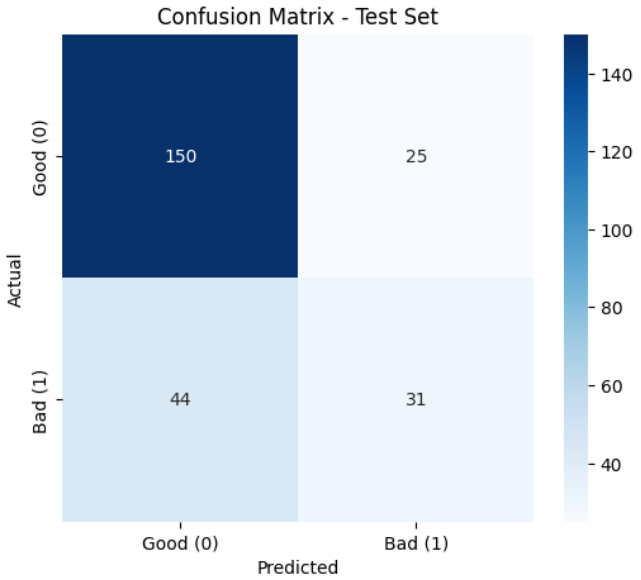

Fig. 3. Confusion matrix for the full logistic regression model (threshold = 0.5).

B. Reduced Logistic Regression

A reduced specification was constructed by removing predictors with weak statistical evidence. This resulted in a model with 19 predictors.
Performance changed only modestly: Test accuracy: 0.75, Bad-class F1-score: 0.50 and Bad-class recall: 0.41
A likelihood-ratio test comparing the reduced and full specifications yielded: LR statistic: 47.64, df difference: 35, p-value: 0.075
This indicates that the additional predictors in the full model did not significantly improve fit at conventional significance levels, despite the increased complexity.

C. Threshold-Adjusted Logistic Model

To improve detection of high-risk clients, the classification threshold was tuned rather than fixed at 0.5. The tuned model achieved: Accuracy: 0.67, Bad-class F1-score: 0.56, Bad-class Recall: 0.71
This improved recall substantially at the cost of lower accuracy, illustrating the trade-off between overall performance and identifying risky borrowers. These results provide empirical evidence for H5, illustrating the expected trade-off between detecting more risky borrowers and maintaining high overall accuracy.

Fig. 4 illustrates how accuracy, recall, and F1-score change as we vary the probability threshold.

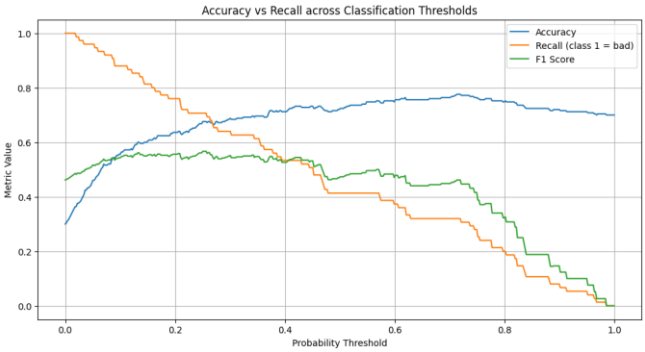

Fig. 4. Effect of decision threshold on accuracy, recall and F1.

D. Lasso-Regularized Logistic Regression

The Lasso logistic model, with the penalty parameter selected via cross-validation, produced: Test accuracy: 0.77, F1-score: 0.53

Several coefficients were shrunk to exactly zero, leaving a smaller subset of relevant predictors. The ROC analysis yielded: AUC = 0.75 (95% CI: 0.680–0.814)

Cross-validated results for Lasso were: Mean accuracy: 0.747, Mean F1-score: 0.51

These were comparable to, and in some cases slightly better than, the unregularized models while using fewer predictors.

E. Bootstrap Results

Bootstrapping the logistic models revealed meaningful variability in both coefficients and performance metrics. While predictors such as checking account status and loan duration remained consistently important across resamples, other coefficients oscillated around zero and frequently changed sign. Accuracy varied by roughly ±0.03 across bootstrap samples, whereas recall for bad credit fluctuated more considerably. This pattern indicates that conclusions based solely on a single fitted model may be overly confident.

To visualize the effect of bootstrapping on the estimated coefficients, we compared the original Lasso coefficients with the mean coefficients obtained across all bootstrap samples. Each point in Fig. 5 corresponds to one predictor, plotted by its original coefficient value on the x-axis and the corresponding bootstrap mean on the y-axis.

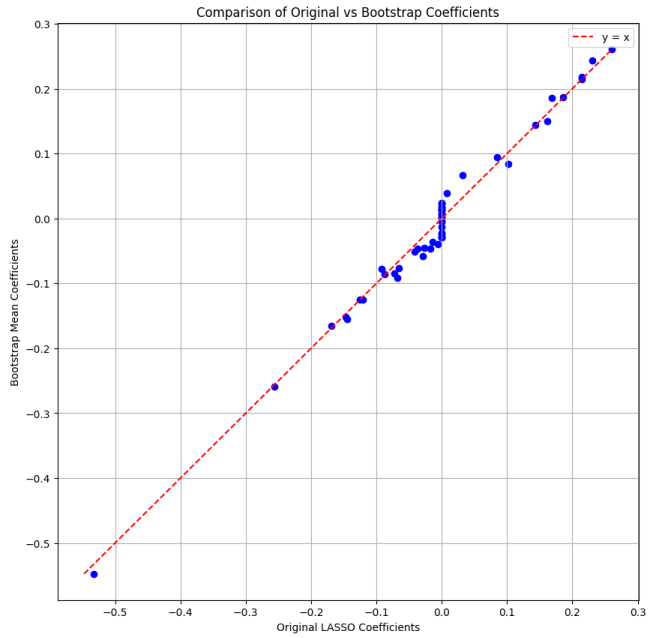

Fig. 5. Comparison of original Lasso coefficients (x-axis) and bootstrap mean coefficients (y-axis).

Most points lie close to the 45-degree line, indicating that the bootstrap averages are very similar to the original estimates and that the Lasso model is relatively stable under resampling. At the same time, the small deviations from the line show that some coefficients are sensitive to the specific sample, which illustrates the sampling variability that is not captured by reporting a single fitted model only.

F. Cross-Validation Comparisons

Cross-validation revealed clear differences across models:

Model	**Mean Accuracy**	**Mean F1 (bad)**	**Notable observation**
Full logistic	0.742	0.517	Unstable coefficients
Reduced logistic	0.771	0.556	Similar performance with fewer predictors
Threshold-tuned logistic	0.694	0.592	Much higher recall, lower accuracy
Lasso logistic	0.747	0.518	Sparse, stable specification

Additionally, a comparison of prediction stability showed: **Lasso:** more consistent across folds, **Standard logistic:** larger fluctuations (mean 0.640, SD 0.015)

Fig. 6 compares the cross-validated accuracy distributions of the Lasso and unregularized models.

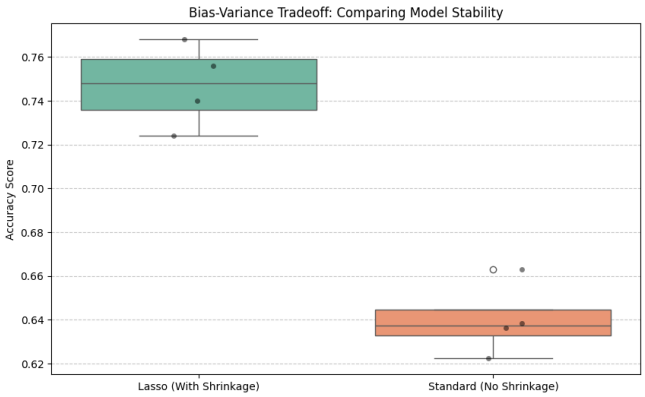

Fig. 6. Cross-validated accuracy distributions for Lasso and standard logistic regression.

G. Predicted Probability Distributions

Density plots of predicted probabilities indicated only partial separation between classes. A large overlap region remained between approximately 0.3 and 0.6, suggesting that part of the classification difficulty may stem from irreducible noise or unobserved predictors.

To understand not only whether a prediction is correct but also how confident the model is, we inspected the distribution of predicted probabilities for good and bad credit applicants. Fig. 7 shows that although good-credit applicants typically receive higher predicted probabilities, the two distributions still overlap substantially.

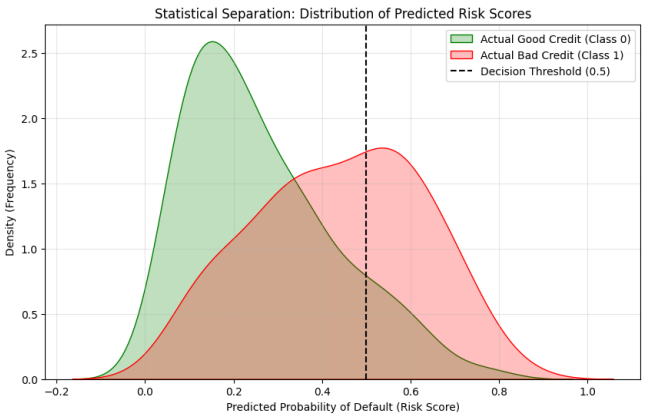

Fig. 7. Distribution of predicted default probabilities for good and bad credit applicants.

This overlap indicates that, even if classification thresholds are adjusted, a subset of applicants remains fundamentally ambiguous, which reflects potential irreducible uncertainty in the data-generating process.

V. Discussion

The results show that interpretable statistical models can achieve performance clearly above a naïve majority-class baseline, but they also highlight the limits of linear decision boundaries in the German Credit dataset. In what follows, we interpret the findings in light of the research question and hypotheses.

The naive classifier would correctly classify approximately 70% of applicants simply by predicting the majority class. All fitted models improved on this baseline: full logistic regression: 0.724 accuracy, reduced logistic regression: 0.752 accuracy, Lasso logistic regression: 0.772 accuracy. These results support H1, indicating that logistic regression captures relevant signal in the predictors. At the same time, gains above the baseline are moderate, suggesting that much of the variance in default outcomes cannot be explained using the available variables alone.

Comparing the full and reduced models revealed that removing statistically weak predictors had almost no negative effect on performance. The likelihood-ratio test suggested that the larger model did not meaningfully improve fit. Similarly, the Lasso model achieved comparable accuracy and F1 while shrinking several coefficients to zero, leading to a simpler and arguably more interpretable specification. These findings are consistent with H2:
"regularization and model reduction can help control overfitting without sacrificing predictive performance."
In practice, this suggests that including many weak predictors may introduce noise rather than useful structure.

Across all models, recall for the bad-credit class remained substantially lower than recall for good-credit cases. Tuning the classification threshold away from 0.5 improved bad-class recall considerably, but at the cost of lower overall accuracy. This behavior confirms H3:
"even well-performing models struggle to reliably detect high-risk clients."
From a decision-making perspective, this implies that performance evaluation should incorporate the asymmetric costs of misclassification, and that threshold tuning may be more important than optimizing a single accuracy value. The threshold analysis directly informs H5. When the decision threshold was reduced to 0.25, recall for bad-credit applicants increased substantially, while overall accuracy declined. This confirms that improving risk detection necessarily increases the number of good applicants incorrectly classified as risky. In realistic lending environments, this trade-off should be addressed using explicit cost or utility functions rather than relying on a fixed threshold.

Bootstrap analysis and cross-validation revealed non-negligible variability in both coefficients and performance metrics. Some predictors were consistently important, while others fluctuated around zero. Test-set results alone therefore risk overstating certainty. This pattern supports H4, showing that conclusions depend on the specific sample and should be interpreted cautiously. Reporting interval estimates and cross-validated metrics is therefore crucial in credit-risk modeling.

Density plots of predicted probabilities showed only partial separation between the two outcome classes. The wide overlap indicates the presence of irreducible error, stemming from either unobserved applicant characteristics, measurement noise, or inherently stochastic behavior.

In other words, even a theoretically "perfectly tuned" logistic model would still misclassify a substantial portion of applicants.

Several limitations should be emphasized:

1. Restricted feature set. Important determinants such as income volatility, debt-to-income ratios, past delinquency severity, or macroeconomic context are absent.
2. Static modeling. The analysis treats applicants independently and does not consider temporal dynamics or repeat borrowing.
3. Linear decision boundary. Non-linear interactions may exist but remain unmodeled.
4. Cost ignorance. Our evaluation treats all errors symmetrically, whereas banks typically apply explicit cost matrices.

Despite these constraints, the study demonstrates that interpretable models supported by regularization, cross-validation, and bootstrap provide transparent and defensible decisions, which is particularly valuable in regulated environments.

Beyond predictive performance, an important advantage of logistic regression is interpretability. In credit scoring, models are frequently audited by regulators and must justify adverse credit decisions to customers. Sparse and transparent models therefore reduce the risk of unfair outcomes, whereas highly complex models may obscure why individuals are denied credit.

Hypothesis	**Supported?**	**Evidence**
H1	Yes	All models outperform majority baseline
H2	Yes	Reduced + Lasso perform similarly with fewer predictors
H3	Yes	Recall lower, threshold tuning trade-offs
H4	Yes (partially)	CV + bootstrap show variability
H5	Yes	Lower threshold increase recall but reduce accuracy

VI. Conclusion

This paper examined the predictive performance and stability of logistic regression models applied to the German Credit dataset. Our objective was to study how regularization, model reduction, and resampling techniques influence accuracy, interpretability, and uncertainty when modeling credit risk.

The results showed that all logistic-based approaches clearly outperformed a naive majority-class baseline, supporting the usefulness of interpretable linear models in this domain. Regularization and variable reduction provided similar or slightly improved performance while yielding more concise models, suggesting that complexity does not automatically translate into improved predictive power. At the same time, recall for the bad-credit class remained notably lower than for the majority class, and tuning the decision threshold revealed clear trade-offs between overall accuracy and risk detection. Threshold analysis further confirmed the expected accuracy–recall trade-off, reinforcing the need for cost-sensitive decision rules rather than relying on a fixed 0.5 cutoff.

Bootstrap resampling and cross-validation highlighted meaningful variability in both coefficients and performance metrics, underscoring the importance of reporting uncertainty rather than relying on a single train–test split. Probability-distribution plots further indicated substantial overlap between good and bad credit cases, pointing to intrinsic limits in what can be predicted from the observed variables.

Overall, the findings suggest that logistic regression, when combined with careful preprocessing, regularization, and resampling, offers a transparent and robust starting point for credit-risk modeling. Future work could extend the analysis by incorporating explicit cost-sensitive evaluation, exploring nonlinear models, or augmenting the feature set with richer behavioral and macroeconomic indicators

VII. References

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